Carbon Dioxide in Brewery Cellars: Ventilation and Alarms

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Lecture notes for the Cellar Air Safety course SKL-2 · Module 3 of 4 · Rev. 4 · issued 8 September 2026

ABSTRACT. These notes accompany the three-hour Module 3 lecture of the Cellar Air Safety course SKL-2, which every employee and contractor working below ground level at Pivovar Doudlevce must complete. They explain how to estimate the carbon dioxide (CO_2) released by our vats and tanks, how to check the ventilation, how long a room stays safe if the fans stop, and how to respond to a Sensilo CD-40 alarm. Rev. 4 replaces Rev. 3 (March 2025). The calculations are planning aids and never override a monitor reading.

1. Scope of these notes

The rules in this module apply to the four below-ground areas of the Doudlevce site: the open fermentation hall F1 (level −1, 1,150 m^3, kept at 8 °C), the sandstone lagering cellars L1 to L6 (level −2, kept at 1–3 °C), the bright beer room B2 (level −1, 640 m^3, kept at 12 °C) and the yeast propagation room Y1 (level −1, 210 m^3).

Module 3 is taught on the first Tuesday of each month in training room 014 next to the cellar entry desk, from 07:30 to 10:30, and ends with a 20-question written test (pass mark 16 of 20). A refresher is required every 24 months; staff whose refresher is overdue by more than 30 days lose their cellar access badge until they attend.

2. Why carbon dioxide is the main cellar hazard

Fresh outdoor air contains about 0.04 % CO_2 by volume. Carbon dioxide is odourless and about 1.5 times as dense as air, so it collects in floor channels, sumps and the stair wells to level −2. For this reason all fixed sensors are mounted 30 cm above the floor, not at breathing height.

For training purposes the brewery uses the following reference levels (site rules, not medical advice):

- 0.5 % (5,000 ppm): highest 8-hour shift average; ventilation design target.
- 1.5 % (15,000 ppm): highest short exposure, 15 minutes at most.
- 3 % (30,000 ppm): headaches, poor judgement.
- Above 8 %: loss of consciousness within minutes.

3. Estimating how much CO_2 a cellar releases

During fermentation, yeast converts sugar into ethanol and CO_2. In the open vats of hall F1 practically all of this gas escapes into the room; in the lagering tanks the spunding valves vent the excess into the cellar. The laboratory measures the apparent extract of every vessel twice a day, so the extract drop is used to estimate the gas release. The mass of CO_2 released in a period is

$$m = k \times V \times \Delta E \tag{1}$$

where m is the mass of CO_2 released (kg), k the release coefficient, fixed at 0.42 kg of CO_2 per hectolitre per degree Plato for all Doudlevce lager worts, V the volume of beer in the vessels concerned (hl) and ΔE the drop in apparent extract over the period (°P).

The hourly gas release rate converts this mass to a volume with the density of CO_2:

$$g = \frac{m}{\rho \times h} \tag{2}$$

where g is the CO_2 release rate (m^3/h), m the mass from formula (1) (kg), ρ the density of CO_2 at the room temperature (kg/m^3) and h the length of the period over which ΔE was measured (hours, normally 24).

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Use these density values: 1.96 kg/m^3 for the lagering cellars L1–L6 (0–4 °C), 1.91 kg/m^3 for hall F1 and room Y1 (5–10 °C) and 1.88 kg/m^3 for the bright beer room B2 (11–16 °C). Rev. 3 used a single value of 1.98 kg/m^3 for every room; calculations made with it before 8 September 2026 slightly underestimate the gas volume and must be redone at the next ventilation review.

> **Example 1 (hall F1 at peak fermentation).** On day 3 of a brew, six open vats of 45 hl each (V = 270 hl) show an extract drop of 2.6 °P in 24 hours. Formula (1) gives m = 0.42 × 270 × 2.6 = 294.84 kg of CO_2 per day. With ρ = 1.91 kg/m^3 and h = 24, formula (2) gives g = 294.84 / (1.91 × 24) = 6.43 m^3 of CO_2 per hour.

> **Example 2 (lagering cellar L4).** Cellar L4 holds twelve horizontal tanks of 150 hl (V = 1,800 hl). During the first week of lagering the extract drops by about 0.15 °P per 24 hours. Then m = 0.42 × 1,800 × 0.15 = 113.4 kg per day and, with ρ = 1.96 kg/m^3, g = 113.4 / (1.96 × 24) = 2.41 m^3/h.

4. Is the ventilation sufficient?

If a room is ventilated with outdoor air at a constant flow, its CO_2 concentration settles at a level where the gas removed by the fans equals the gas released by the beer. Rearranging this balance and adding the site safety factor gives the minimum fresh-air flow:

$$Q_{min} = \frac{S \times g}{c_{lim} - c_{out}} \tag{3}$$

where Q_{min} is the minimum fresh-air flow (m^3/h), S the safety factor (1.5 for all cellar rooms; it covers uneven mixing near the floor), g the CO_2 release rate from formula (2) (m^3/h), c_{lim} the design concentration of 0.005 (0.5 % by volume) and c_{out} the outdoor concentration, taken as 0.0004 (0.04 %). Enter concentrations as volume fractions.

> **Example 3 (hall F1).** With g = 6.43 m^3/h from Example 1, Q_{min} = 1.5 × 6.43 / (0.005 − 0.0004) = 1.5 × 6.43 / 0.0046 ≈ 2,097 m^3/h. Hall F1 has two extract fans, VF-1 and VF-2, each rated at 1,400 m^3/h at floor level. Both fans together (2,800 m^3/h) are sufficient at peak fermentation; one fan alone is not. If either fan is out of service, no more than four vats may be at days 2 to 4 of fermentation at the same time.

For cellar L4, Example 2 gives Q_{min} = 1.5 × 2.41 / 0.0046 ≈ 786 m^3/h, well within the 1,200 m^3/h of the shared level −2 fan VL-3. The installed fan capacities of the other rooms are: B2, 900 m^3/h (fan VB-1); Y1, 450 m^3/h (fan VY-1); L1–L3 together, 1,600 m^3/h (fan VL-1); L5 and L6 together, 1,000 m^3/h (fan VL-4).

5. How long a room stays safe if the fans stop

When the ventilation stops, the CO_2 released by the beer stays in the room and the concentration rises. Ignoring leakage through doors and stair wells (which makes the answer conservative), the time until a given concentration is reached is

$$t = \frac{V_{room} \times \left(c_A - c_0 \right)}{g} \tag{4}$$

where t is the time after the fans stop (hours), V_{room} the free air volume of the room (m^3), c_A the concentration of interest as a volume fraction (for example 0.015 for the main alarm), c_0 the concentration when the fans stopped (volume fraction, read from the monitor) and g the CO_2 release rate (m^3/h).

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> **Example 4 (hall F1, fan power cut).** The monitor reads 0.10 % ($c_0 = 0.001$) when both F1 fans stop at peak fermentation ($g = 6.43$ m³/h). The main alarm level of 1.5 % is reached after $t = 1{,}150 \times (0.015 - 0.001) / 6.43 \approx 2.5$ hours, and the pre-alarm level of 0.5 % after $1{,}150 \times (0.005 - 0.001) / 6.43 \approx 0.72$ hours, i.e. about 43 minutes.

The free air volumes used for formula (4) are: F1, 1,150 m³; B2, 640 m³; Y1, 210 m³; L1, 780 m³; L2, 760 m³; L3, 740 m³; L4, 820 m³; L5, 690 m³; and L6, 610 m³. These volumes exclude tanks and vats (survey of May 2026).

6. The Sensilo CD-40 monitoring system

Every below-ground room has at least two fixed Sensilo CD-40 infrared CO_2 sensors (hall F1 has four), all mounted 30 cm above the floor and wired to the panel at the cellar entry desk on level 0. The thresholds are:

- **Pre-alarm** at 0.5 % (5,000 ppm): yellow lamp, intermittent tone. Until 31 August 2026 the pre-alarm was set at 1.0 %; all sensors were reprogrammed to 0.5 % in the first week of September 2026.
- **Main alarm** at 1.5 % (15,000 ppm): red beacon, continuous siren, fans switch to boost mode automatically.
- **Evacuation alarm** at 3.0 % (30,000 ppm): red beacon and two-tone siren in all rooms on the same level; access doors unlock for exit and lock for entry.

Each fixed sensor receives a bump test with 2.0 % test gas every 30 days by the night shift electrician and a full two-point calibration every 6 months by the service contractor. A sensor that fails a bump test is marked with a red tag; until it is replaced, the room may only be entered with a portable monitor and a second person waiting at the door.

7. What to do when an alarm sounds

The response depends on the alarm level. On a **pre-alarm**, safely stop the task within five minutes, check the fans and report the reading to the shift leader by radio. On a **main alarm** or an **evacuation alarm**, follow these steps in order:

1. Stop work immediately.
2. Leave the room by the nearest exit, keeping your head as high as possible; never go down into a lower area, a sump or the level −2 stair well to leave.
3. Report to the assembly point at the cellar entry desk on level 0 and give your name to the desk attendant, who checks it against the entry log.
4. Do not re-enter to rescue a colleague. Call the site emergency number 4444 and tell the attendant who is missing and where they were last seen. Only the trained rescue team with breathing apparatus may go in.
5. Re-entry is allowed only after the shift leader has confirmed with a portable monitor, from the doorway, that the reading is below 0.5 % and the fans have run in boost mode for at least 20 minutes.

Fan bypass. No one may switch a cellar fan off or put a sensor in maintenance mode while staff are in the room. Bypassing the fan interlock for maintenance requires a written permit signed by the Head Cellar Master (Ing. Petr Kalous) or, in his absence, the duty process engineer, and is limited to one shift.

References

1. Pivovar Doudlevce a.s. (2026). *Cellar ventilation survey, levels −1 and −2*. Internal report PD-TECH-118, May 2026.
2. Kalous, P. and Hronková, J. (2024). Extract drop as a predictor of gas release in open fermentation. *Bulletin of the West Bohemian Brewers' Circle*, 31, 44–52.

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